Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/15/2024
PI Name:	Hirsa, Amir Ph.D.		
Project Title:	Amyloid Fibril Formation in Microgravity via Ring-Sheared Drop		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	12/31/2023
No. of Post Docs:	3	**No. of PhD Degrees:**	6
No. of PhD Candidates:	11	**No. of Master' Degrees:**	7
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	15
No. of Bachelor's Candidates:	24	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gorti, Sridhar	**Contact Phone:**	256 544-0158
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/31/2023 per NSSC information (Ed., 6/5/23) NOTE: End date changed to 05/31/2023 per NSSC information (Ed., 10/5/22) NOTE: End date changed to 9/30/2022 per continuation grant 80NSSC20K1726 issued, including title change now reflected in above Title, per S. Gorti/MSFC (Ed., 1/27/21) NOTE: Extended to 9/30/2020 per PI and NSSC information (Ed., 6/4/18)		
Key Personnel Changes/Previous PI:	June 2020 report: no changes		
COI Name (Institution):	Lopez, Juan Ph.D. (Arizona State University)		
Grant/Contract No.:	80NSSC20K1726 ; NNX13AQ22G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note January 2021: Continuation grant issued with slightly different title, Amyloid Fibril Formation in Microgravity via Ring-Sheared Drop 80NSSC20K1726 (formerly titled Amyloid Fibril Formation in Microgravity: Distinguishing Interfacial and Flow Effects NNX13AQ22G). Information combined with original grant NNX13AQ22G, per NASA Marshall Space Flight Center management] Formation of amyloid fibrils is widely studied because of their role in neurodegenerative diseases, including Alzheimer's. The fibrillization process is characterized by a nucleation step followed by fibril growth, much like the steps that occur during protein crystallization. Aside from the general interest in macromolecular biophysics, recent evidence shows the relevance of amyloid fibril formation in animal models exposed to cosmic radiation, with obvious implications for interplanetary travel. A preponderance of evidence shows that agitating protein solutions can significantly accelerate amyloid fibril formation; however, the mechanisms by which the presence of bubbles and flow independently affect fibrillization are not known. The situation is further complicated because we do not yet have mechanistic models that can quantify the influence of natural convection due to gravity on the fibrillization process. We know that fluid interfaces (e.g., air-water interface) induce amyloid fibrils when they are sheared, but the influence of container walls is not fully understood. Here, we aim to distinguish the influence of flow and fluid interfaces on amyloid formation by removing other effects, including solid walls and buoyancy effects. Formation of protein amyloid fibrils provides a model system for studying biomolecular transport phenomena, where the influence of flow can be altogether negated by the elimination of gravity. Furthermore, the experiments can be performed aboard the International Space Station (ISS) with essentially no solid container, since surface tension can provide liquid containment. Thus, the free surface can be readily observed (e.g., with high magnification objectives), manipulated (e.g., can be sheared or left otherwise undisturbed), and modified (e.g., by spreading a monomolecular oil film). First, we will perform experiments at 1g on a small droplet to ascertain the effect of forced convection on a liquid surface. Next, we will generalize our very successful computational model in annular systems to the spherical geometry of liquids at microgravity. We will design the space-based experiments based on the results from the ground experiments and numerical predictions. Finally, we will perform experiments aboard ISS on large-scale drops in an experiment referred to as the Ring-Sheared Drop (RSD).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	When proteins are perturbed from their native state, they lose their native confirmation (folding) and then can bond with other proteins, forming an aggregate. This is the reason why the transparent fluid in an egg becomes a white solid when cooked. Amyloid fibrils are a unique type of protein aggregate that occur naturally in the body. We study them because amyloid plaques in the brain are associated with many neurodegenerative disorders, including Alzheimer's. Fibrils are very intriguing because they are highly ordered and self-assemble. When heated, insulin molecules, for example, denature and bond with other denatured molecules (or assists in denaturing the other molecules). These form a nucleation site, and from there the fibril elongates as more molecule bond into what is called a "beta sheet" structure. The primary mechanism for activation of protein in the body is not yet understood, but it is likely to be a combination of flow friction, aka "shear stress," and interfacial effects that native proteins are exposed to. Aside from their central role in neurodegenerative diseases such as Parkinson's and Alzheimer's, formation of amyloid structures in proteins plays a role in other diseases including diabetes.
Task Progress:	Interaction of proteins with fluid interfaces constitutes a principal component of a molecule's energetic landscape. The effects of fluid interfaces on proteins are ubiquitous throughout physiology and the environment, as well as many industries, including medicine and biotechnology. Study of fluid interfaces can be uniquely accomplished in microgravity, as surface tension becomes dominant in the absence of gravity, allowing the production of large and stable air-liquid interfaces. The ring-sheared drop "RSD" is a microgravity biochemical reactor comprised of a liquid drop pinned between two contact rings. One ring rotates to transmit shear, via surface shear viscosity, and subsequently produce mixing in the bulk fluid by flow inertia. In the RSD, surface tension provides containment. This apparatus, operating aboard the International Space Station (ISS), was used to study the fibrillization of human insulin in response to fluid shear with minimal impact from solid boundaries. Such studies followed experiments within the ground analog of the RSD, the knife-edge (surface) viscometer "KEV," a device that retains the interfacial shear driving and bulk fluid inertia mixing of the RSD, with the caveat that a glass dish is necessary for containment under the effects of gravity. Human insulin was utilized in these investigations as a model amyloid protein, related to diseases such as Alzheimer's, Parkinson's, infectious prions, and type 2 diabetes, to study hydrodynamically-induced fibrillization kinetics in the presence of air-liquid interfaces, minimizing effects of non-physiological solid boundaries. Physical processes affecting proteins at fluid interfaces include adsorption, structural conformation, and kinetic processes, all manifesting as a result of hydrodynamics. Molecular transport determines the microenvironment of a protein, subsequently altering the protein's structure to obtain the lowest free energy, leading to changes in protein function. During RSD experiments aboard the ISS studying amyloidogenesis of human insulin, upon extraction of a 4 mg/mL drop after steady shearing for 3.5 days, a structure was observed at the air-liquid surface. This membranous, wrinkled feature indicated the presence of an interfacial protein gel with mechanical properties distinct from those of the bulk fluid. Observations of protein gelation at the air-liquid interface within the RSD during ISS experiments motivated ground investigation of insulin gelation using the KEV. This KEV experiment was designed for direct quantitative observation of insulin's air-liquid interface gelation by particle tracking velocimetry "PTV" of hollow glass microspheres introduced to the air-liquid interface. Independent variables of shear rate and protein concentration were investigated for their impact on the dependent variable of interfacial gelation with the hypothesis that if shear rate and/or protein concentration increase, then the rate of insulin gelation at the air-liquid interface will increase as a result of transport-mediated amyloidogenesis.
Bibliography Type:	Description: (Last Updated: 04/15/2026)

Abstracts for Journals and Proceedings	Adan JA, McMackin PM, Riley FP, Adam CE, Underhill PT, Hirsa AH. "Protein gelation at the air-liquid interface in microgravity and on Earth." ISSRDC 2024 (International Space Station R&D Conference 2024). Boston, MA, July 30-August 1, 2024. Abstracts. ISSRDC 2024 (International Space Station R&D Conference 2024). Boston, MA, July 30-August 1, 2024. , Jul-2024